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Enhancing Fault Mapping in Complex Structure with Applications of Ultra-Deep Azimuthal Resistivity Imaging

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Abstract

Heavily faulted reservoirs require clear understanding of their structures for optimal development and reservoir management. Relying on seismic data has been the main enabler for identifying faulted regions, however, seismic data resolution is impacted by the throws (shifts) of the faults, resulting in many faults being undetected. This study discusses the use of Ultra-Deep Resistivity (UDAR) image in accurate detection of faults, and how these results can gap the uncertainties of fault mapping.

A candidate well, located within a heavily faulted area, has been selected for the utilization of UDAR imaging. The well provides a unique opportunity for mapping the congestion of faults and their potential extension towards the surrounding areas. The objectives of utilizing UDAR imaging include providing a comprehensive insight on the complex structure of the area, and assisting the geo- steering in a challenging trajectory crossing several faults. The UDAR imaging results have been benchmarked with resistivity image log and seismic data to identify the areas of uncertainties in faults identification from each data source.

The obtained results from the UDAR image log added substantial information and data points for a better understanding of the area. The UDAR results identified several unexpected faults that were not detected previously by seismic data. The tool also estimated the throws (shifts) of each detected fault clearly, while they were not captured by seismic due to limitations in seismic resolution for detecting minor faults with small throws. The picked faults from UDAR have been confirmed with conventional resistivity image tool, as a benchmark approach to confirm their accuracy. Besides fault detection, the tool provided valuable insights on the extension of several fractures, ultimately assisting the lower completion design and compartmentalization of the lateral. Combining the data obtained from seismic, resistivity images, and UDAR image provided a comprehensive view of the complex structure of the area, ultimately assisting the planning of wells, and optimal development in the future.

Combining structural data obtained from several sources resulted in a better delineation of complex structures and faulted regions. The paper discusses the process of structural delineation starting from seismic, resistivity images, and finally complimenting the data with UDAR image logs to gap uncertainties. It provides a clear reference for areas with similar challenges by identifying the values and limits of each data source.

Introduction

In the modern energy industry, accurate fault detection is of paramount importance for oil and gas professionals to effectively understand their reservoirs and to facilitate better decision making. Traditional methods for fault detection include the use of 3D seismic surveys and resistivity image logs. However, these methods have inherent limitations. Seismic data, while valuable for large-scale geological mapping, often fails to detect smaller faults due to resolution constraints. On the other hand, resistivity image logs offer high-resolution images of the wellbore environment, helping to identify fractures and bed boundaries. However, these logs are limited by their shallow depth of investigation, typically up to a few inches, which restricts their ability to provide detailed information about fault displacement.

To address these challenges, this paper presents an example whereby an ultra-deep azimuthal resistivity (UDAR) tool was deployed while drilling a well, with the objective of detecting both the presence and displacement of subsurface faults. The UDAR log, initially designed for geosteering, offers a significantly deeper depth of investigation compared to traditional resistivity tools, making it more suitable for detecting faults that may not be resolved by seismic data. This tool's ability to provide insights into larger-scale structural features could significantly enhance fault detection capabilities in oil and gas exploration and development.

The objective of this paper is to compare the fault detection capabilities of the UDAR log with those of traditional 3D seismic surveys and shallow resistivity image logs. By evaluating the effectiveness of the UDAR log in identifying faults and its limitations, this research aims to provide insights into its potential as a complementary tool in geological mapping and reservoir management strategies. The implications of this study could lead to improved well placement, enhanced understanding of the geological formation, and more confident decision-making based on a greater amount of data.

Theory of LWD Tools and Methods

The resistivity image logging-while-drilling (LWD) tool works by sending highly focused currents through the drilling mud into the formation. With its multi-button configuration and azimuthal acquisition system, the tool can generate high-resolution borehole images around the wellbore while drilling (Al-Balooshi et al, 2017). These images, which can reach as high as 0.2-inch vertical resolution, are very effective in revealing formation features such as fractures, vugs, and bed boundaries. However, the depth of investigation is very limited—typically up to 3 inches.

Even when the wellbore intersects a fault, it is often difficult to distinguish it from natural fractures using image data alone. In many cases, a systematic analysis of bedding dips before and after the interpreted feature is required to confirm the presence of a fault. Due to this limited depth of investigation, even when a fault is identified, the resistivity image tool cannot determine the displacement across the fault. This limitation makes it challenging to geosteer effectively in reservoirs with complex faulting and structural geometry.

The UDAR tool employs a fundamentally different approach to reservoir mapping compared to conventional resistivity imaging tools. Rather than injecting focused currents into the formation, the UDAR tool's transmitter emits electromagnetic (EM) waves at multiple low frequencies. These signals propagate through the formation and are subsequently received and measured by multiple receivers along the bottom-hole assembly (BHA).

Several design features enable the UDAR tool to achieve a significantly greater depth of investigation—often exceeding 100 feet from the wellbore—than standard LWD tools. The first key factor is the physical separation between the transmitter and the receiver modules. By placing receivers at larger spacings along the BHA, the tool can detect deeper formation features. This configuration also allows for a combination of high-resolution mapping relatively in the proximity of the wellbore and ultra-deep mapping farther away (Fig. 1).

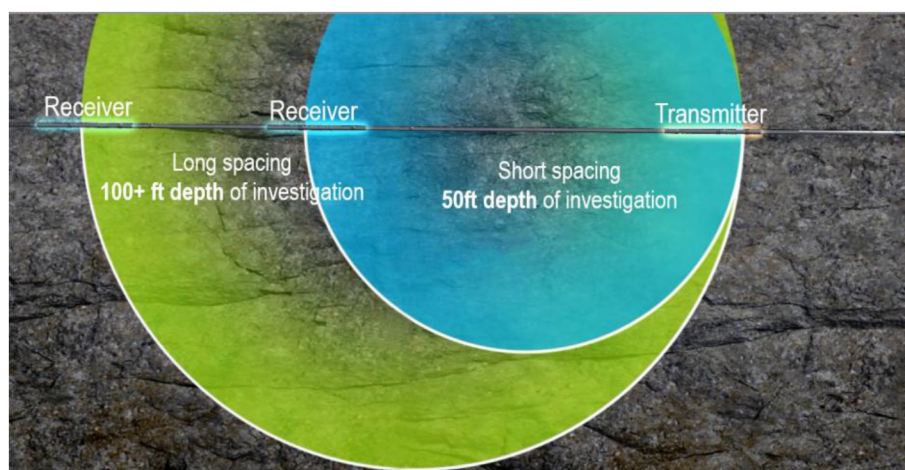


Figure 1—UDAR tool transmitter and receivers placement and the corresponding depth of investigation

A second critical factor is the use of lower-frequency EM waves. In general, the lower the transmission frequency, the deeper the waves can penetrate into the formation. UDAR tools typically operate at frequencies 50 to 100 times lower than those used in conventional resistivity tools, thus extending their depth of investigation. Additionally, the tool can transmit across a range of frequencies, providing a spectrum of measurements. This feature enhances adaptability across various geological settings, from low-resistivity overburden to high-resistivity hydrocarbon-bearing zones. The third major innovation is the azimuthal sensitivity enabled by the design of the transmitter and receiver antennas. Traditional propagation resistivity tools use only axial antennas, resulting in measurements that lack directional sensitivity. In contrast, the UDAR tool incorporates tilted antennas at various angles. As the tool rotates with the BHA, it transmits and receives EM waves from multiple orientations, allowing it to resolve resistivity variations in all azimuthal directions. This design enables full 360-degree imaging of the formation around the wellbore and supports 3D reservoir mapping (Seydoux et al, 2014).

The UDAR tool typically generates a large set of real-time measurements due to its multi-receiver, multi-frequency, and multi-azimuthal antenna configuration. These measurements cannot be directly interpreted using conventional methods. Instead, a dedicated inversion algorithm processes the data to produce an intuitive visual representation: the inverted resistivity map, or inversion canvas. This map uses a color scale to depict resistivity variations above and below the wellbore, enabling interpreters to identify formation boundaries and resistivity contrasts.

The presence of a fault can be clearly identifiable on the inversion canvas because of the discontinuity of resistivity along layers mapped by the tool. Moreover, the displacement along the fault could be estimated by measuring the offset of a resistive formation layer. This has been described by several authors, including Viandante et al (2015) and Goodin et al (2024).

Because the UDAR tool can map formation boundaries within the reservoir, it is highly valuable for geosteering in horizontal wells. This is particularly important in the case of our extended-reach drilling (ERD) well. In ERD wells, the large lateral step-out distance introduces significant uncertainty in the target's position, both laterally and vertically, especially when seismic-related uncertainties are considered. These uncertainties can make it challenging to land the well precisely as planned.

By using the UDAR tool, the geosteering team can clearly visualize formation boundaries in real time as the well progresses. This allows for accurate adjustments to the trajectory inclination based on the UDAR inversion results, enabling the well to enter the target zone at the intended angle. In contrast, without the UDAR tool, an unexpected fault throw that shifts the target zone just a few feet deeper could result in missing the reservoir and drilling unnecessary intervals in the overlying formation.

Application and Modelling for High Resistivity Environment

For the subject well under discussion, in addition to mapping faults and reservoir structures, there was also a requirement to detect and characterize fractures within the reservoir. This necessitated the deployment of both the UDAR tool and the conventional resistivity imaging tool in the bottom-hole assembly (BHA), introducing additional complexity in BHA design and measurement-while-drilling (MWD) telemetry management. After consultations with various teams, the final BHA configuration was determined as shown in Fig. 2.



Figure 2—BHA configuration for the job

The UDAR transmitter was placed as the first tool above the rotary steerable system (RSS) to ensure proximity to the bit and to minimize the sensor offset between the UDAR inversion data and the bit position. Immediately above the UDAR transmitter, the resistivity imaging tool was positioned. This tool provides both laterolog resistivity curves and high-resolution borehole images at various depths of investigation. Placing it close to the bit ensures that resistivity images are captured before significant mud invasion occurs. Additionally, this tool functions as a physical spacer between the UDAR transmitter and the first UDAR receiver.

Following the first UDAR receiver, the BHA includes the primary MWD tool, which supplies power and mud-pulse telemetry for both the UDAR and imaging tools. This is followed by the second UDAR receiver. The two-receiver configuration of the UDAR system provides sufficient spacing for deep resistivity measurements, while still enabling high-resolution imaging near the wellbore via the first receiver.

Because density and porosity measurements were also required, a dual-BHA setup was adopted. The density and porosity logging tools were put in the second BHA, positioned above the primary BHA described above. These two BHAs operate independently, meaning they do not share power sources or telemetry bandwidth. This flexible configuration allowed the integration of all required tools without compromising operational performance.

In carbonate reservoirs, hydrocarbon-bearing zones typically exhibit very high resistivity. However, other features within carbonate formations—such as tight zones and anhydrite—can also display similarly high resistivity. For boundary detection at a distance from the wellbore, the UDAR tool depends on the conductivity contrast between adjacent layers. When formation resistivity is high, the conductivity contrast between layers is often low, making it more difficult for the UDAR tool to distinguish boundaries.

Furthermore, in ultra-high resistivity environments (greater than 1000 Ohm·m), the formation induced electromagnetic (EM) signals is very weak to start with, resulting in weak signal reception at the receiver antennas. To address these challenges, three strategies can be employed:

1. Enhance signal-to-noise ratio (SNR) at the receivers by using highly sensitive antennas.
2. Increase the transmitter firing power to improve the likelihood of signals reaching the receivers.
3. Transmit EM waves at higher frequencies, which tend to perform better in high-resistivity formations.

Fig. 3 shows the modeling results for different resistivity contrast scenarios. In all cases, the resistivity of the formation layer in which the tool is positioned is set to 1000 Ohm·m, corresponding to a conductivity of 0.001 S/m. The resistivities of the approaching (underlying) formation are set to 200 Ohm·m, 500 Ohm·m, and 800 Ohm·m for scenarios (a), (b), and (c), respectively.

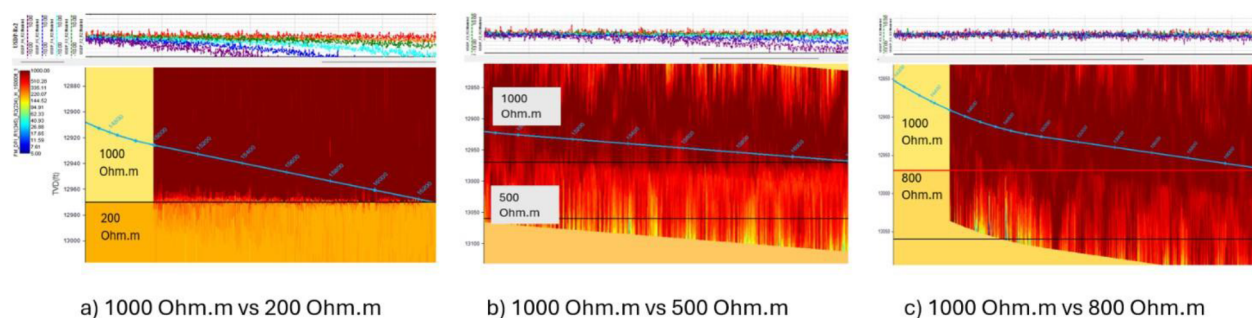


Figure 3—UDR modeling result for high resistivity environments

The top log tracks display a set of measurements most sensitive to formation boundaries. In scenario (a), a clear response is observed as the tool approaches the boundary with the 200 Ohm·m formation. The logs begin to deflect significantly as the tool nears the boundary, and the inversion clearly maps the interface from as far as 40 ft true vertical depth (TVD) below the wellbore.

In scenario (b), the log deflection begins closer to the boundary, and while the response is less pronounced than in scenario (a), it remains visible and interpretable. The inversion result shows a blurrier but still recognizable boundary, providing a reasonably confident interpretation.

In scenario (c), where the resistivity contrast is smallest (1000 Ohm·m vs. 800 Ohm·m), the logs show minimal deflection, and the boundary is not clearly mapped in the inversion, even when the wellbore is in close proximity. This highlights the challenge of detecting low-contrast boundaries in high-resistivity environments.

Results from Real-Time Execution

The execution of this challenging extended-reach drilling (ERD) section was successful, and it achieved a new record as the longest horizontal section drilled in this area.

Results from High-Resolution Resistivity Imaging Tool

First, we examine the results from the high-resolution resistivity imaging tool discussed earlier. Fig. 4 displays some selected intervals using the standard resistivity image colormap, where brighter colors indicate higher resistivity and darker colors indicate lower resistivity. Two visual representations of the same data are shown: the left panel uses a static colormap, while the right panel uses a dynamic colormap to enhance contrast and highlight subtle features.

With a resolution of 0.2 inches, the high-resolution resistivity image enables detailed identification of various formation characteristics. For example, on Fig. 4.a, distinct karst features were observed—indicated by a sinusoidal purple dip pick on a dark resistive anomaly—which may be associated with the mud losses encountered during drilling. On Fig. 4.b, a minor fault was interpreted based on the abrupt change in bedding dip and resistivity patterns before and after the feature. However, estimating the fault throw using resistivity images alone remains difficult due to the limited depth of investigation.

Another valuable capability of this imaging tool is its ability to identify fractures—both resistive and conductive types (Fig. 4.f). In particular, conductive fractures are of significant interest due to their potential association with water pathways.

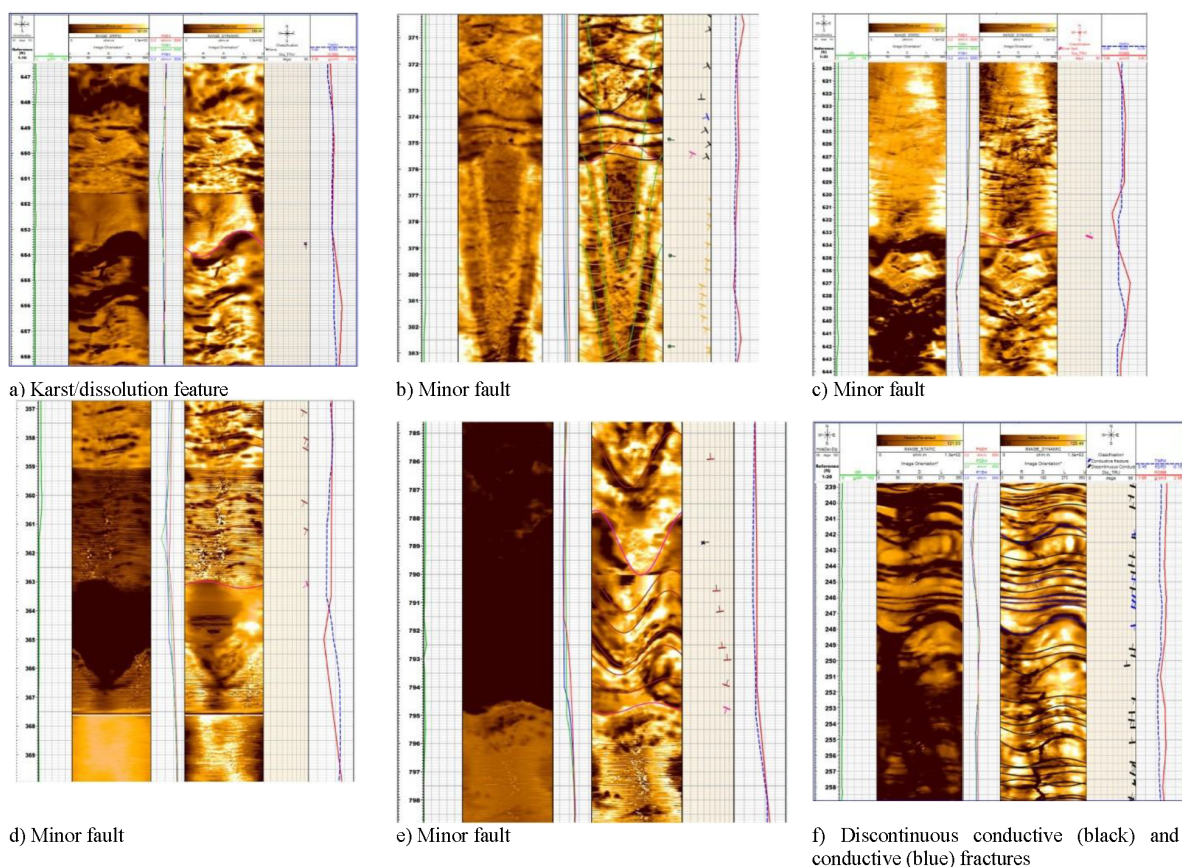


Figure 4—Selected high-resolution resistivity images, with interpreted dips.

Fig. 5 presents all interpreted bedding dips from this section in a 3D view. It is evident that this portion of the well is highly fractured, featuring karst structures and several minor or micro-faults. Some of these faults caused noticeable displacements, shifting the well trajectory across different stratigraphic zones. However, these shifts are not apparent when examining the resistivity images alone.

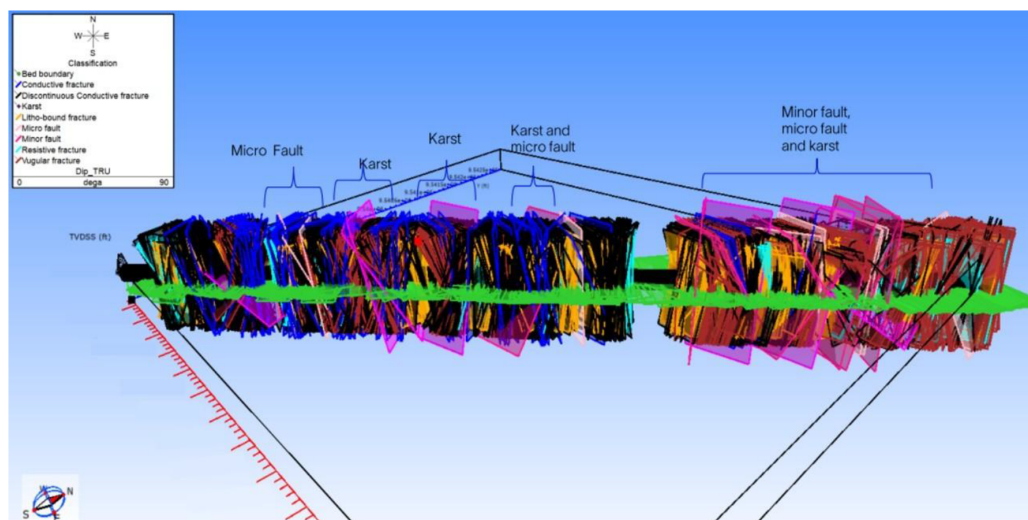


Figure 5—Overview of the manually picked features from the high-resolution resistivity image. Here the pink dip planes are faults identified from image.

Results from UDAR Tool

Fig. 6 shows the first part of the final UDAR result for the well, alongside corresponding log and image data. The top track displays the high-resolution resistivity image discussed earlier. The second track from the top contains the density and neutron porosity logs. The third track presents resistivity logs with a scale ranging from 0.2 to 20,000 Ohm·m: the orange curve represents the at-bit resistivity, while the blue curve corresponds to the 40-inch spacing phase shift propagation resistivity. The final track shows the gamma ray (GR) log, the average rate of penetration (ROP5), and the near-bit inclination (PDINCL). Below the log tracks, the UDAR inversion result is displayed on a resistivity canvas, overlaid on the interpreted structural model. In the inversion canvas, warm colors indicate high resistivity and cool colors indicate low resistivity, following the colormap scale shown on the left (5 to 1000 Ohm·m).

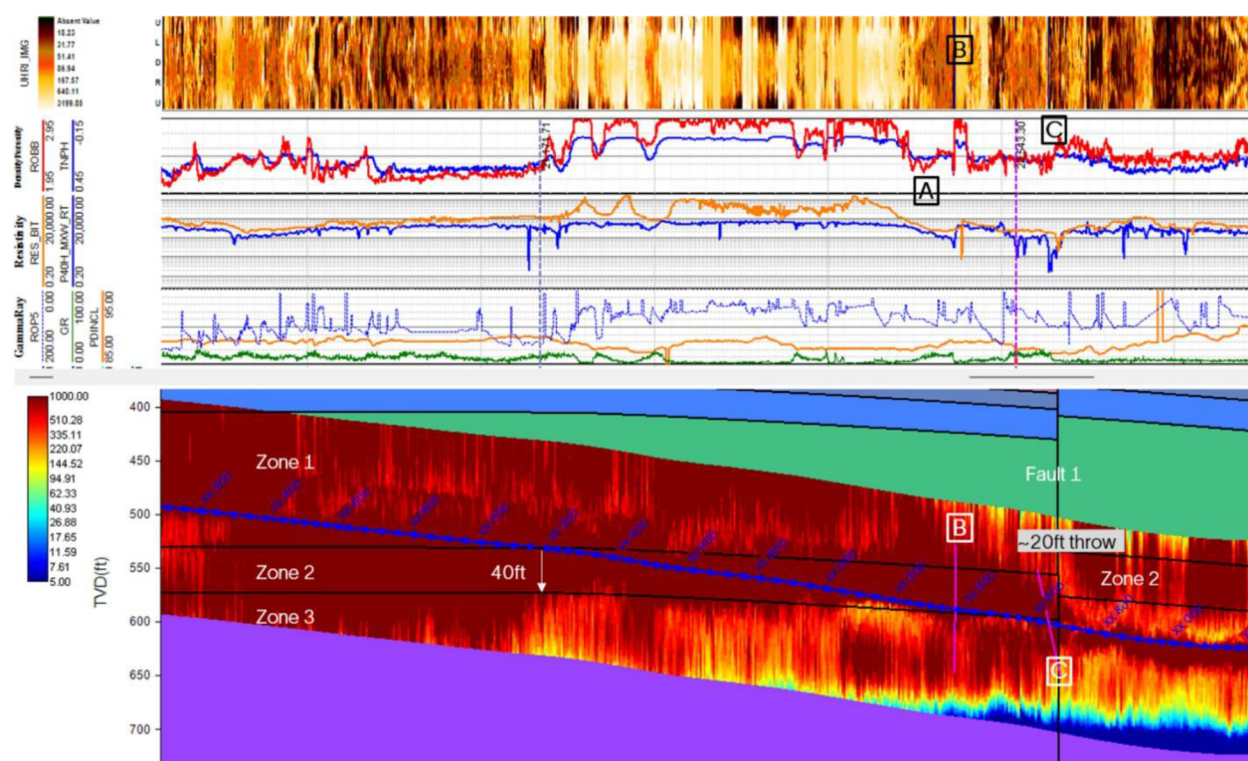


Figure 6—Well LWD image, logs, and UDR inversion results for the landing interval of interest.

As shown in Fig. 6, while drilling in Zone 1, the tool had difficulty identifying the boundary between Zones 1 and 2 due to their similarly high resistivities—both exceeding 1000 Ohm·m. However, upon entering Zone 2, a very tight anhydrite layer with resistivity occasionally exceeding 10,000 Ohm·m, the UDAR inversion began to detect a boundary with a more conductive formation approximately 40 feet true vertical depth (TVD) below the wellbore—indicating the top of Zone 3.

This information was instrumental in guiding the landing into Zone 3, which was the target reservoir. Based on the inversion data, the decision was made to reduce the inclination slightly to accelerate the trajectory through the anhydrite and enter the target zone. The well successfully entered Zone 3 (Point A, Fig. 6), confirmed by the drop in both resistivity and density logs.

Shortly after entering Zone 3—around 150 ft MD into the target—the well encountered a fault and briefly entered a tight, resistive interval (Point B, Fig. 4) for approximately 37 ft MD, before re-entering Zone 3. This fault is clearly visible on the resistivity image (Fig. 4.d), but was not detected in the UDAR inversion due to its small throw.

Approximately 250 ft MD beyond the previous feature, the well encountered another fault—clearly visible on both the UDAR inversion (Point C in Fig. 6) and the resistivity image (Fig. 4.c). This fault brought

the wellbore slightly deeper within Zone 3, as indicated in the UDAR inversion display. However, the fault was not apparent on conventional LWD logs, since the well remained within Zone 3 both before and after crossing the fault, resulting in minimal change in the log responses. The direction and displacement of the fault were inferred from the UDAR inversion, which indicated an upthrown fault with an estimated throw of approximately 20 feet.

After crossing Fault 1, the well continued drilling through the upper part of Zone 3, maintaining an estimated distance of approximately 20 to 35 ft above the base of Zone 2. Following 740 ft of drilling in this zone, the well intersected another fault (Fault 2 in Fig. 7), which shifted the wellbore upward in the stratigraphy, placing it into the lower part of Zone 2. This shift was evident from the UDAR inversion and further confirmed by increases in both density and resistivity logs.

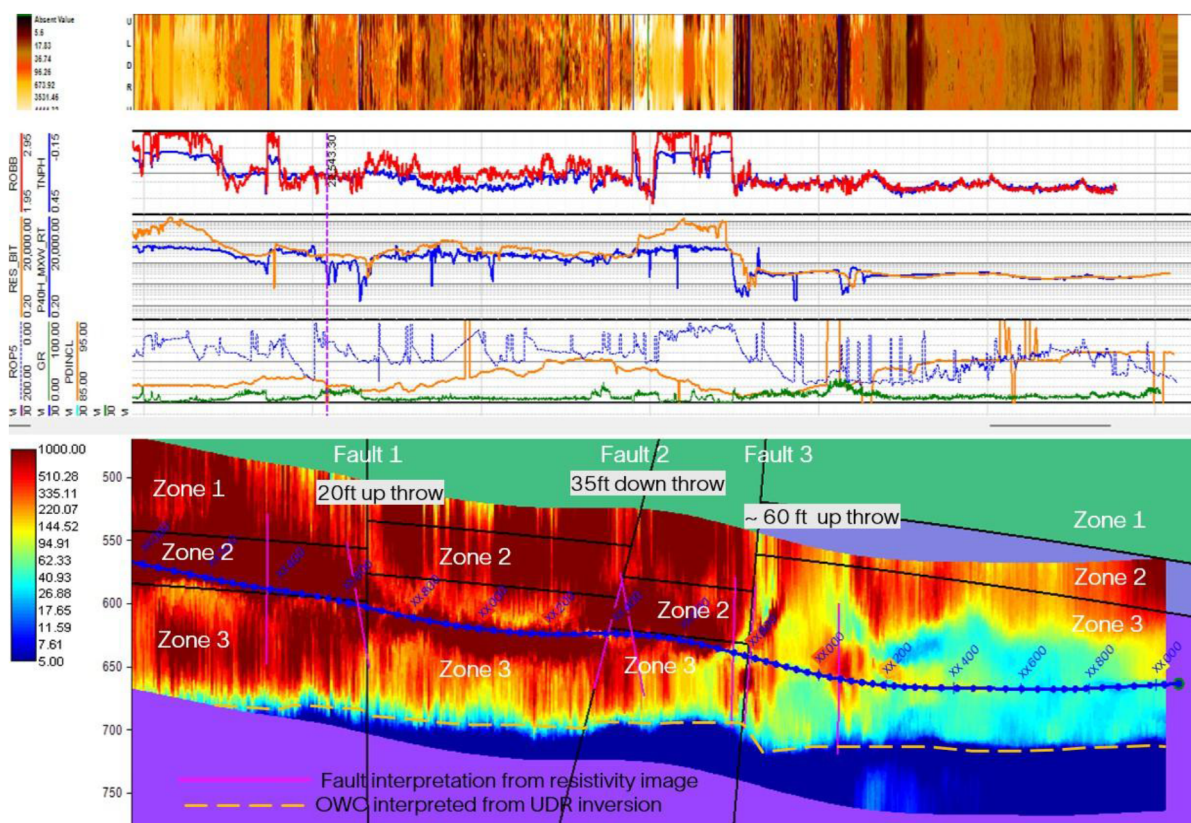


Figure 7—Well image and logs and UDAR inversion result towards TD of the well. Faults identified from resistivity image are shown as purple solid lines. OWC interpreted from UDAR inversion was shown as yellow dash lines.

However, the UDAR inversion also indicated that the wellbore was positioned very close to the boundary between Zone-2 and Zone-3. Based on this insight, the decision was made to drop the inclination in order to re-enter Zone 3. The well successfully drilled back into Zone 3 after approximately 370 ft MD beyond the point of intersection with Fault 2.

Shortly after re-entering Zone 3, the well encountered another fault zone (Fault 3 in Fig. 7). This time, the resistivity dropped to 50 Ohm·m or lower. Correlation with offset well data suggested that the well had reached the lower part of Zone 3. This interpretation was soon confirmed by the UDAR inversion, which mapped the high-resistivity Zone-2 approximately 75 ft above the wellbore. Fault 3 was also clearly visible on the high-resolution resistivity image (Fig. 4.e). In this final fault block, the well was drilled with a near-horizontal trajectory, and total depth (TD) was called after 1,260 ft of drilling—successfully staying 100% within the target Zone 3.

Another notable feature observed in the UDAR inversion is a conductive zone (dark blue) located near the bottom of the inversion canvas. This highly conductive layer, with an estimated resistivity of approximately 3 Ohm·m, was interpreted as the potential oil-water contact (OWC). The interpreted OWC is highlighted as a yellow dashed line in Fig. 7. This feature was first detected when the well initially entered Zone-3, appearing approximately 85 ft below the wellbore. Notably, the apparent depth of the OWC varied slightly before and after crossing Fault 3, suggesting structural displacement or stratigraphic variation across the fault.

These insights are highly valuable for reservoir engineers, particularly in informing completion design and optimizing the production strategy for this well.

Discussions

The well discussed in this paper penetrated four different oil-bearing formations, all of which are carbonate. These carbonates are a mixture between limestone and dolomite rock types. Despite their similar composition, the UDAR tool was not effective in detecting the first three formations. To understand this in greater detail, it is important to recognize that the fourth oil-bearing zone, which the UDAR tool was able to log successfully, is likely composed of facies with a higher clay content. As an example, some of the high clay content facies found include mud-supported packstone and wackestone. These facies typically translate into lower formation resistivity, which explains why the UDAR tool was able to map this fourth formation effectively.

This idea can be expanded further by considering how the subject well was geosteered. While geosteering, it is imperative to stay close to the top of the oil-bearing formation. This was challenging given the highly faulted nature of the formation, while also considering the dogleg limitations that must be obliged to. Ultimately, the benefit of the UDAR tool while geosteering across highly fault formations can be seen in Fig. 8. Despite the presence of faults, the well was geosteered with a high degree of accuracy, meaning that a minimal amount of non-pay formation was encountered. The net-to-gross ratio between oil-bearing porous rock and tight rock is approximately 91%. Given the challenging environment of this field, this is a notably high percentage, part of which can be attributed to the UDAR technology.

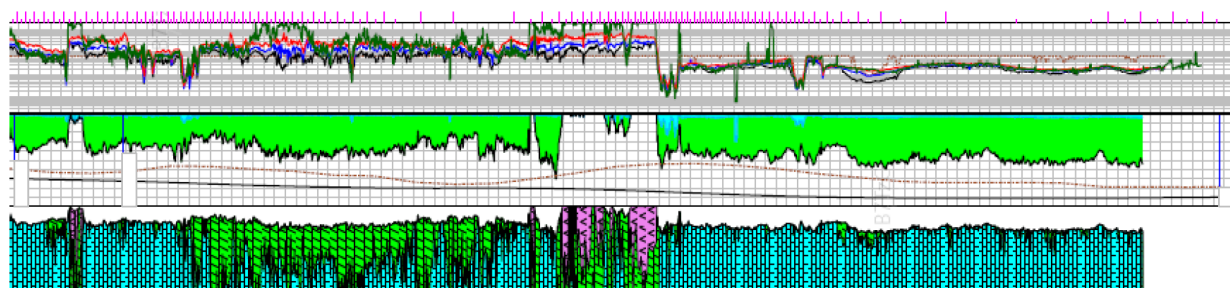


Figure 8—Part of the formation analysis log from the well discussed in this paper alongside the resistivity curves and lithological overview.

Conclusion

This paper presented a successful example in which the UDAR tool was used to map faults in a high-resistivity environment, in combination with the high-resolution LWD resistivity imaging tool to identify fractures and confirming the faults. The integration of these tools provided the operator with valuable real-time insights into fault location and displacement during drilling this extended reach well. This capability helped mitigate risks associated with structural uncertainties and facilitated informed decision-making—not only for geosteering, but also for completion design and future well planning. However, a key limitation of the technology lies in its dependence on resistivity contrast. In some carbonate formations with extremely

high resistivity and minimal contrast between layers, the tool may face challenges in accurately mapping structural features. It is important to note that the subject well discussed in this paper benefited from the combination of traditional seismic data, shallow resistivity image log, and the UDAR tool. Each of their respective data comes with certain limitations. However, when coupled together, they are able to provide powerful new insights for better delineation of the highly faulted environment.

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